5G CELL SCANNER

"Cyberspectrum is the best spectrum"

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University Status: Undergraduate Student

Programming Languages Mastered: C, Python

Synopsis

This project aims to develop a high-quality 5G cell scanner using GNU Radio and an SDR to passively observe and decode cellular signals. Unlike existing scanners, which often stop at basic detection, this tool will extract key broadcast information such as the Master Information Block (MIB) and System Information Block 1 (SIB1) to provide detailed insights into detected cells.

By leveraging GNU Radio's capabilities, this project will push the boundaries of real-time cellular monitoring and coverage mapping, offering researchers and engineers a powerful, open-source tool for 5G network analysis. Additionally, implementing multi-band scanning will improve network analysis efficiency, making the tool more useful for real-world applications. The scanner will provide detailed insights into network configurations, allowing users to analyze synchronization signals, extract system information, and log critical network parameters. Current open-source solutions, such as free5GRAN, offer similar capabilities but are limited in certain aspects, such as real-time performance, ease of use, and integration with SDRs. The goal is to provide an efficient and detailed method for 5G network exploration while maintaining extensibility for future enhancements.

Background Information

Wireless Communication Basics

5G operates in **Sub-6 GHz** and **mmWave** bands. It uses **QPSK, 16-QAM**, and **64-QAM** for higher data rates but requires good **SNR**. **OFDMA** improves efficiency for both uplink and downlink. Signals are affected by **fading**, **Doppler shifts**, and **multipath interference**.

- **Frequency Bands**: **Sub-6 GHz** for coverage, **mmWave** for speed.
- **Modulation**: **QPSK, 16-QAM, 64-QAM**.
- **Multiplexing**: **OFDMA** for uplink and downlink.
- **Channel Effects**: **Fading**, **Doppler shifts**, **multipath interference**.

Digital Signal Processing (DSP)

FFT analyzes signals in the frequency domain. **Filtering** reduces **noise** and **interference**. **Synchronization** ensures correct signal timing and alignment, while **error correction** (e.g., **Polar Codes** and **LDPC**) enhances decoding accuracy.

- **FFT**: For frequency analysis and locating **SSBs**.
- **Filtering**: Reduces **noise** and **interference**.
- **Synchronization**: Ensures signal alignment.

5G NR Protocols

The **SSB** is the first signal detected, containing **PSS**, **SSS**, and **PBCH**. The **PBCH** carries the **MIB**, which provides **cell configuration**. **SIBs** (e.g., **SIB1**) carry network parameters like **access info**.

- **SSB**: Contains **PSS, SSS, PBCH**.
- **PBCH**: Encodes **MIB** with **cell info**.
- **SIBs**: Carry network config details like **access info**.

Software-Defined Radio (SDR)

SDRs use **IQ sampling** to represent signals, and **GNU Radio** provides customizable blocks for signal processing.

- **IQ Sampling**: Represents signals in **In-phase (I)** and **Quadrature (Q)** components.
- **GNU Radio**: Used for building **custom DSP pipelines**.

Implementation Plan

Phase 1: Python-Based DSP Pipeline (Weeks 1-4)

- Implement 5G synchronization (SSB detection) in Python.
- Implement PBCH decoding to extract MIB information.
- Extend decoding to SIB1 extraction and initial visualization.
- Validate and test DSP pipeline using recorded SDR data.

Deliverable: A working Python prototype that detects 5G cells and extracts MIB/SIB1.

Phase 2: Integration with GNU Radio (Weeks 5-8)

- Convert the Python DSP pipeline into GNU Radio blocks.
- Implement MIB/SIB1 decoding as a GNU Radio flowgraph or OOT module.
- Develop real-time visualization and logging features.
- Test with live SDR input and refine GNU Radio implementation.

Deliverable: A functional GNU Radio-based 5G cell scanner.

Phase 3: Optimization & Finalization (Weeks 9-10)

- Optimize performance and explore multi-band scanning (if feasible).
- Conduct final testing, debugging, and refinement.
- Complete documentation and prepare the code for submission.

Deliverable: Fully working 5G cell scanner with documentation, visualization, and logging.

Final Submission & Wrap-Up

- Submit the final implementation and documentation.
- Share results with the GNU Radio community and gather feedback.
- Explore potential future improvements beyond GSoC.

Technologies & Tools

- **GNU Radio** – Framework for SDR-based signal processing.
- **Python (NumPy, SciPy, Sionna, Toolkit5G, Py3GPP)** – DSP implementation and 5G-specific processing.
- **SDR (USRP, RTL-SDR, BladeRF, LimeSDR)** – Hardware for receiving 5G signals.
- **MATLAB References** – Guides for implementing synchronization and decoding algorithms.

Challenges & Considerations

- **Complexity of 5G Signal Processing**: Adapting MATLAB algorithms to Python and GNU Radio.
- **Computational Performance**: Ensuring real-time processing efficiency.
- **Multi-Band Scanning**: Parallelizing signal processing for improved coverage.
- **SDR Limitations**: Hardware constraints in capturing and processing 5G signals.

Future Work & Extensions

- **Enhancing Multi-Band Scanning**: Improving efficiency in scanning multiple frequencies simultaneously.
- **Real-Time Mobile Scanning**: Implementing GPS tagging to generate real-time coverage maps based on SDR location.
- **Further Downlink Decoding**: Exploring beyond MIB/SIB1 to decode additional system information messages.

Benefits to the Community

- Expands GNU Radio's application to cellular network analysis.
- Provides an open source 5G monitoring tool for researchers and engineers.
- Enhances understanding of 5G signal structure and network characteristics.
- Helps improve coverage mapping and network optimization strategies.

Additional Information

I have engaged with the GNU Radio community through mailing lists, forums, and discussions related to this project. This has helped me gain insights into the project scope, technical challenges, and available resources.

I would like to work with **Michael Petry** as a mentor, given his expertise and guidance on this.

I also had a productive call with my mentor, Michael Petry, to discuss the scope, technical direction, and refinement of my GSoC proposal for the 5G Cell Scanner project.

I have taken a close look at the approach used by our community member Daniel Estévez in PBCH decoding(https://destevez.net/2024/12/5g-nr-pbch/) and tried this myself. This has given me valuable insights into 5G synchronization and decoding techniques, which I plan to incorporate into this project.

I have also examined how 5G cells work by reviewing the GNU Radio Conference 2021 talk by Aaron Madsen(https://www.youtube.com/watch?v=-xh-9w7usW0&ab_channel=GNURadio).

Additionally, I have referred to free5GRAN's open-source repository (https://github.com/free5G/free5GRAN) for insights into existing 5G signal processing methodologies.

I have also developed a Colab notebook(https://colab.research.google.com/drive/10l3gbFTkCH0sDfWx77h0ZJdmkGhohKrw?usp=sharing) based on my research, incorporating concepts from Daniel's work. The notebook includes the code I have written and the sample data I used for initial testing.

I have studied the MATLAB-based approach to 5G cell search and MIB/SIB1 recovery as described in this MathWorks guide, which provides insights into synchronization, broadcast channel decoding, and implementation strategies that inform my project development.

I went through the GNU Radio tutorials and gained hands-on experience in building and running flowgraphs, which helped me understand its architecture and signal processing capabilities.

University Credit for GSoC Project: No, this is not in connection with my university work, and I am not getting credit for the GSoC project.

Why I Am a Good Fit for This Project

I believe I am a great fit for this project because it aligns perfectly with my passion for wireless communications, signal processing, and open-source development. My journey into 5G and SDR started with curiosity diving into research papers, exploring community discussions, and experimenting with real-world 5G signal samples. I have not just studied the theory but have actively implemented and tested my own approaches. I thrive on problem-solving and collaboration.